COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	08/26/2026	**DATE SUBMITTED:**	08/29/2026
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ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (*CPE232_yourname*). Issue the command git pull. When prompted, enter the correct passphrase or password. Describe what happened when you issue this command. Did something happen? Why?
   - When I ran the git pull command, the terminal displayed already up to date which means that there were no new changes to download because my local repository was already updated with the latest files from the github.

```
charles-vence-espinas@workstation:~$ cd CPE212_CharlesVenceEspinas
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas$ git pull
Already up to date.
```

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): *ansible-playbook --ask-become-pass install_apache.yml*. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
all:
  children:

      ubuntu:
        hosts:
          server1:
            ansible_host: 192.168.56.105
            os_type: Ubuntu

          server2:
            ansible_host: 192.168.56.106
            os_type: Ubuntu

      centos:
        hosts:
          centOS:
            ansible_host: 192.168.56.102
            os_type: CentOS
```

```
      linuxservers:
        children:
          ubuntu:
            vars:
              ansible_user: charles-vence-espinas
          centos:
            vars:
              ansible_user: charles-vence-espinas
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] ******************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [update repository index] *************************************************
changed: [server1]
changed: [server2]

TASK [install apache2 package] *************************************************
ok: [server2]
ok: [server1]

TASK [add PHP support for apache] **********************************************
ok: [server2]
ok: [server1]
```

```
PLAY [centos] ******************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [update repository index] *************************************************
ok: [centOS]

TASK [install httpd package] ***************************************************
ok: [centOS]

TASK [add PHP support for apache] **********************************************
ok: [centOS]

TASK [start httpd service] *****************************************************
ok: [centOS]

PLAY RECAP *********************************************************************
centOS                     : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

3. Edit the *install_apache.yml* file and insert the lines shown below.

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```yaml
---
- hosts: ubuntu
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes

    - name: install apache2 package
      apt:
        name: apache2

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php

- hosts: centos
  become: true
  tasks:

    - name: update repository index
      dnf:
        update_cache: yes

    - name: install httpd package
      dnf:
        name: httpd
```

```yaml
    - name: add PHP support for apache
      dnf:
        name: php
        state: present

    - name: start httpd service
      service:
        name: httpd
        state: started
        enabled: true
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [ubuntu] ******************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [update repository index] *************************************************
changed: [server1]
changed: [server2]

TASK [install apache2 package] *************************************************
ok: [server2]
ok: [server1]

TASK [add PHP support for apache] **********************************************
ok: [server2]
ok: [server1]
```

- In this part, I use the playbook that I used in the HOA 4 and that is because I fixed the CentOS error in that HOA and this is why I used this because it outputs the same as this step in task 1 in which it will install apache, php and update the cache which exactly the same as the previous HOA which is HOA 4 and the instruction stated that used the previous playbook.

  If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

  - name: update repository index
  apt:
    update_cache: yes
  when: ansible_distribution in [“Debian”, “Ubuntu]

  Note: This will work also if you try. Notice the changes are highlighted.

```
PLAY [centos] ******************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ar
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [update repository index] *************************************************
ok: [centOS]

TASK [install httpd package] ***************************************************
ok: [centOS]

TASK [add PHP support for apache] **********************************************
ok: [centOS]

TASK [start httpd service] *****************************************************
ok: [centOS]

PLAY RECAP *********************************************************************
centOS                     : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored
```

4. Edit the *install_apache.yml* file and insert the lines shown below.

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
      stae: latest
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache2 package
    dnf:
      name: httpd
      state: latest
    when: ansible_distribution == "CentOS"

  - name: add PHP support for apache
    dnf:
      name: php
      state: latest
    when: ansible_distribution == "CentOS"
```

```
---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: update repository index
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

```
    - name: install apache2 package
      dnf:
        name: httpd
        state: latest
      when: ansible_distribution == "CentOS"

    - name: add PHP support for apache
      dnf:
        name: php
        state: latest
      when: ansible_distribution == "CentOS"
```

Make sure to save and exit.

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
```

```
TASK [update repository index] *************************************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:9:13

7       apt:
8         update_cache: yes
9       when: ansible_distribution == "Ubuntu"
              ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
changed: [server2]
changed: [server1]
```

```
TASK [install apache2 package] *************************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:15:13

13        name: apache2
14        state: latest
15      when: ansible_distribution == "Ubuntu"
              ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
ok: [server2]
ok: [server1]
```

```
TASK [add PHP support for apache] **********************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:21:13

19        name: libapache2-mod-php
20        state: latest
21      when: ansible_distribution == "Ubuntu"
            ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
ok: [server1]
ok: [server2]
```

```
TASK [update repository index] *************************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:26:13

24       dnf:
25         update_cache: yes
26       when: ansible_distribution == "CentOS"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [server1]
skipping: [server2]
ok: [centOS]
```

```
TASK [install apache2 package] *************************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:32:13

30         name: httpd
31         state: latest
32       when: ansible_distribution == "CentOS"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [server1]
skipping: [server2]
changed: [centOS]
```

```
TASK [add PHP support for apache] **********************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:38:13

36         name: php
37         state: latest
38       when: ansible_distribution == "CentOS"
               ^ column 13

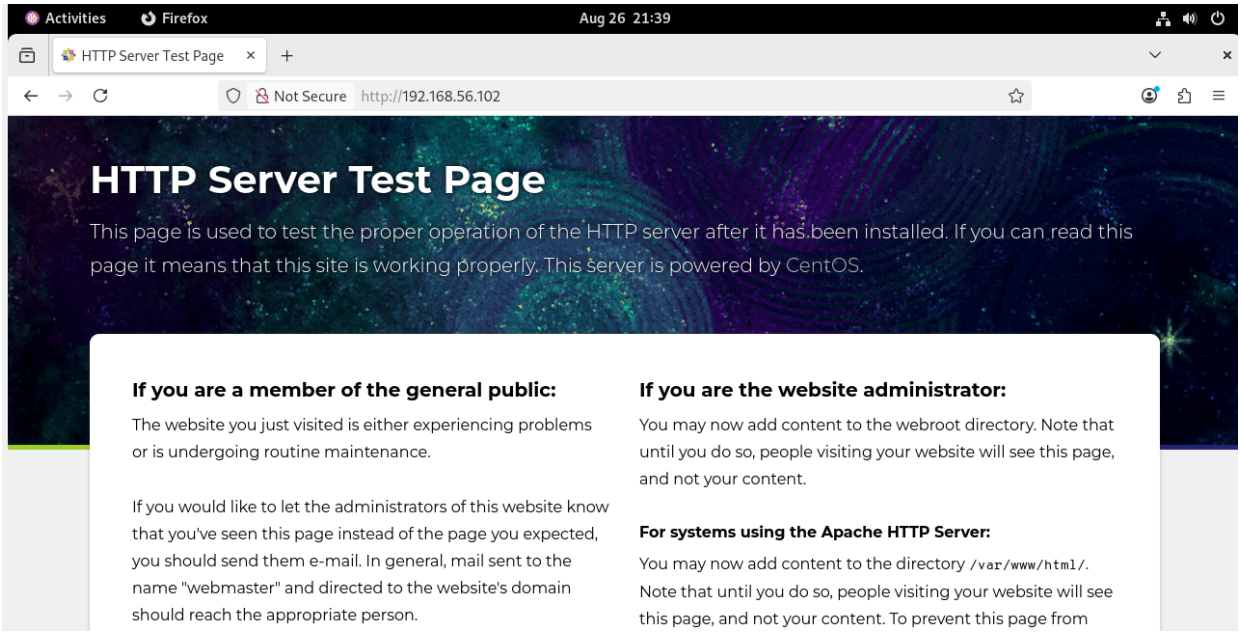
Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [server1]
skipping: [server2]
ok: [centOS]

PLAY RECAP *********************************************************************************************************
centOS                     : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

- In this part, I modify my playbook and align to this HOA's instruction and then the output of the playbook is it ran on all the hosts. Then the apt tasks ran on server 1 and server 2 and it skipped on centOS. The dnf tasks then ran on to centOS to install the httpd and it skipped the server 1 and server 2 which is the ubuntu.

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in the CentOS is not yet active. Thus, you need to activate it first.



5.1 To activate, go to the CentOS VM terminal and enter the following:

systemctl status httpd

The result of this command tells you that the service is inactive.

```
[charles-vence-espinas@centOS ~]$ systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Wed 2026-08-26 21:25:40 PST; 15min ago
       Docs: man:httpd.service(8)
   Main PID: 5575 (httpd)
     Status: "Total requests: 14; Idle/Busy workers 100/0;Requests/sec: 0.0155; Bytes served/sec: 5.9KB/sec"
      Tasks: 177 (limit: 5559)
     Memory: 7.9M (peak: 14.6M)
        CPU: 787ms
     CGroup: /system.slice/httpd.service
             ├─5575 /usr/sbin/httpd -DFOREGROUND
             ├─5583 /usr/sbin/httpd -DFOREGROUND
             ├─5585 /usr/sbin/httpd -DFOREGROUND
             ├─5586 /usr/sbin/httpd -DFOREGROUND
             └─5587 /usr/sbin/httpd -DFOREGROUND

Aug 26 21:25:39 centOS systemd[1]: Starting The Apache HTTP Server...
Aug 26 21:25:39 centOS httpd[5575]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::a00:27ff:fef2:beb9%enp>
Aug 26 21:25:40 centOS httpd[5575]: Server configured, listening on: port 80
Aug 26 21:25:40 centOS systemd[1]: Started The Apache HTTP Server.
lines 1-22/22 (END)
```

5.2 Issue the following command to start the service:

sudo systemctl start httpd

(When prompted, enter the sudo password)

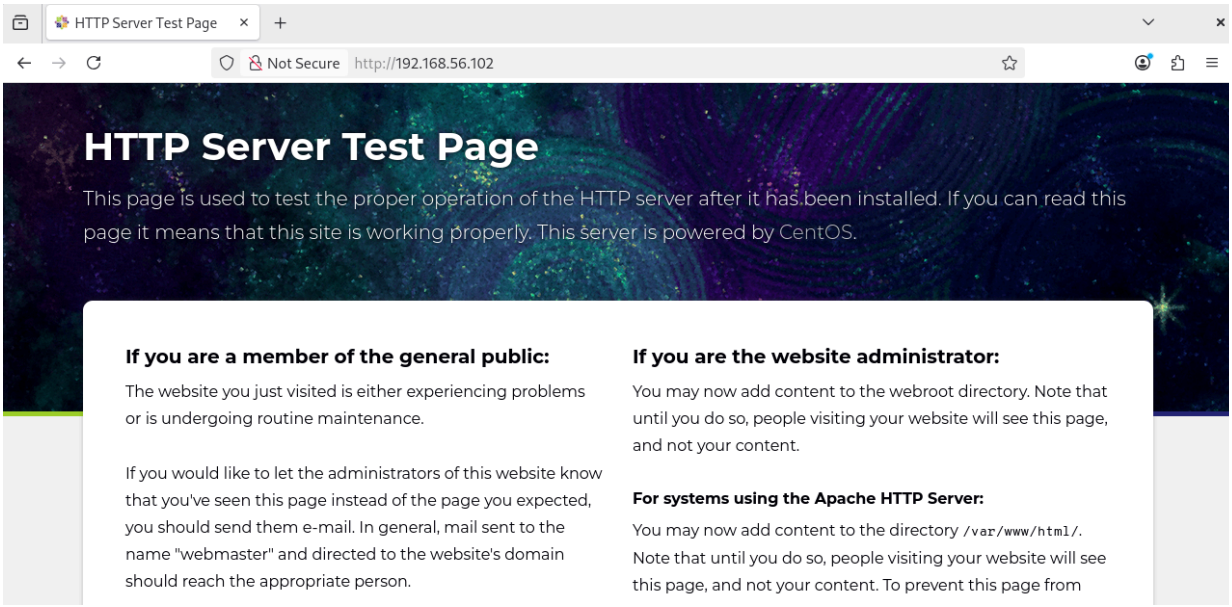
sudo firewall-cmd --add-port=80/tcp

(The result should be a success)

```
[charles-vence-espinas@centOS ~]$ sudo firewall-cmd --add-port=80/tcp
success
[charles-vence-espinas@centOS ~]$
```

- As you can see in the browser, I successfully activated the httpd in step 5 since I already do that in previous HOA which is HOA 4. So the status is it is already enabled.

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



- It is successful.

Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook *install_apache.yml*. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```
---
- hosts: all
  become: true
  tasks:

  - name: update repository index Ubuntu
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: update repository index for CentOS
    dnf:
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install apache and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

```
---
- hosts: all
  become: true
  tasks:

    - name: update repository index Ubuntu
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 packages and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: update repository index for CentOS
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

```
  - name: install apache2 packages and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
```

```
TASK [update repository index Ubuntu] *****************************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:9:13

7       apt:
8         update_cache: yes
9       when: ansible_distribution == "Ubuntu"
              ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
changed: [server1]
changed: [server2]
```

```
TASK [install apache2 packages and php packages for Ubuntu] *******************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:17:13

15          - libapache2-mod-php
16        state: latest
17      when: ansible_distribution == "Ubuntu"
              ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
ok: [server1]
ok: [server2]
```

```
TASK [update repository index for CentOS] ***************************************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:22:13

20       dnf:
21         update_cache: yes
22       when: ansible_distribution == "CentOS"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [server1]
skipping: [server2]
ok: [centOS]
```

```
TASK [install apache2 packages and php packages for CentOS] *********************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:30:13

28           - php
29         state: latest
30       when: ansible_distribution == "CentOS"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [server1]
skipping: [server2]
ok: [centOS]

PLAY RECAP ***********************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

- The playbook finished faster because the task which is the installation is combined together. Putting multiple packages in the list under apt and dnf reduced the number of the tasks while still using the when conditions for each operating systems.

2. Edit the playbook *install_apache.yml* again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidated everything in just 2 plays. This can be done by removing the update repository play and putting the command *update_cache: yes* below the command *state: latest*. See below for reference:

```
---
- hosts: all
  become: true
  tasks:

    - name: install apache2 and php packages for Ubuntu
      apt:
       name:
         - apache2
         - libapache2-mod-php
       state: latest
       update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

```
---
- hosts: all
  become: true
  tasks:

    - name: install apache2 packages and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 packages and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
      when: ansible_distribution == "CentOS"
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
```

```
TASK [install apache2 packages and php packages for Ubuntu] ********************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:13:13

11         state: latest
12         update_cache: yes
13       when: ansible_distribution == "Ubuntu"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [centOS]
ok: [server1]
ok: [server2]
```

```
TASK [install apache2 packages and php packages for CentOS] ********************************************************
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injected a
fter the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:22:13

20         state: latest
21         update_cache: yes
22       when: ansible_distribution == "CentOS"
               ^ column 13

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

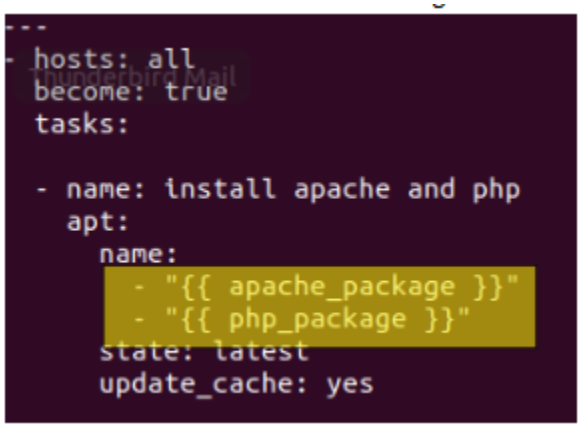
skipping: [server1]
skipping: [server2]
ok: [centOS]

PLAY RECAP *********************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- The playbook ran without the separate tasks to update the package repositories. Using the update_cache yes inside the package tasks updated the package list before installing the package.

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the apache_package and php_package are variables. The names are arbitrary, which means we can choose different names. We also take out the line when: ansible_distribution. Edit the playbook *install_apache.yml* again and make sure to follow

the below image. Make sure to save the file and exit.




```
---
- hosts: all
  become: true
  tasks:

    - name: install apache packages and php packages
      apt:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
```

```
TASK [install apache packages and php packages] *********************************************************************
[ERROR]: Task failed: Finalization of task args for 'ansible.builtin.apt' failed: Error while resolving value for 'name'
: 'apache_package' is undefined

Task failed.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:6:7

4   tasks:
5
6     - name: install apache packages and php packages
        ^ column 7

<<< caused by >>>

Finalization of task args for 'ansible.builtin.apt' failed.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:7:7

5
6     - name: install apache packages and php packages
7       apt:
        ^ column 7

<<< caused by >>>

Error while resolving value for 'name'.
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:9:11
```

```
7       apt:
8         name:
9           - "{{ apache_package }}"
            ^ column 11

<<< caused by >>>

'apache_package' is undefined
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/install_apache.yml:9:13

7       apt:
8         name:
9           - "{{ apache_package }}"
              ^ column 13

fatal: [server1]: FAILED! => {"changed": false, "msg": "Task failed: Finalization of task args for 'ansible.builtin.apt'
 failed: Error while resolving value for 'name': 'apache_package' is undefined"}
fatal: [server2]: FAILED! => {"changed": false, "msg": "Task failed: Finalization of task args for 'ansible.builtin.apt'
 failed: Error while resolving value for 'name': 'apache_package' is undefined"}
fatal: [centOS]: FAILED! => {"changed": false, "msg": "Task failed: Finalization of task args for 'ansible.builtin.apt'
failed: Error while resolving value for 'name': 'apache_package' is undefined"}

PLAY RECAP **********************************************************************************************************
centOS                     : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

- The playbook failed because of install apache packages and php packages tasks. This happened because ansible could not find the variables {{ apache_package }} and {{ php_package }} because they were not defined yet.

4. Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the *inventory* file and follow the below configuration:

```
192.168.56.120 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.121 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.122 apache_package=httpd php_package=php
```

Make sure to save the *inventory* file and exit.

```
all:
  children:

    ubuntu:
      hosts:
        server1:
          ansible_host: 192.168.56.105
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php

        server2:
          ansible_host: 192.168.56.106
          os_type: Ubuntu
          apache_package: apache2
          php_package: libapache2-mod-php

    centos:
      hosts:
        centOS:
          ansible_host: 192.168.56.102
          os_type: CentOS
          apache_package: httpd
          php_package: php
```

```
  linuxservers:
    children:
      ubuntu:
        vars:
          ansible_user: charles-vence-espinas
      centos:
        vars:
          ansible_user: charles-vence-espinas
```

Finally, we still have one more thing to change in our *install_apache.yml* file. In task 2.3, you may notice that the package is assign as apt, which will not run in CentOS. Replace the *apt* with *package*. Package is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use *apt*, and for CentOS it will automatically use *dnf*. Make sure to save the file and exit. For more details about the ansible package, you may refer to this

documentation: [ansible.builtin.package – Generic OS package manager — Ansible Documentation]

```
---
- hosts: all
  become: true
  tasks:

    - name: install apache packages and php packages
      package:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

Run *ansible-playbook --ask-become-pass install_apache.yml* and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook --ask-become-pass install_a
pache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation o
f another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-
core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of
another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [install apache packages and php packages] *******************************************************************
ok: [server1]
ok: [server2]
ok: [centOS]
```

```
PLAY RECAP ********************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

- The playbook finally ran successfully on all hosts with ok and no errors. By defining the package variables in inventory.yaml, it gave the ansible the correct package names. Also, by changing the apt to the package module also allowed the ansible to automatically use the correct package manager for ubuntu and centOS.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:

Create a playbook that could do the previous tasks in Red Hat OS.

/etc/hosts

```
  GNU nano 5.6.1                                                    /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.56.105 server1
192.168.56.106 server2
192.168.56.104 server3
```

Inventory for CentOS

```
  GNU nano 5.6.1                                    inventory_forcentOS.yaml
all:
  children:
    ubuntu:
      hosts:
        server1:
          ansible_host: server1
          apache_package: apache2
          php_package: libapache2-mod-php
        server2:
          ansible_host: server2
          apache_package: apache2
          php_package: libapache2-mod-php
        server3:
          ansible_host: server3
          apache_package: apache2
          php_package: libapache2-mod-php
      vars:
        ansible_user: charles-vence-espinas
```

Playbook

```
  GNU nano 5.6.1                                              install_apache.yml
---
- hosts: all
  become: true
  tasks:

    - name: install apache packages and php packages
      package:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

Output

```
[charles-vence-espinas@centOS ansible-lab]$ ansible-playbook -i inventory_forcentOS.yaml install_apache.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
ok: [server3]
ok: [server1]
ok: [server2]

TASK [install apache packages and php packages] *********************************************************************************************************************
ok: [server1]
ok: [server2]
changed: [server3]

PLAY RECAP *********************************************************************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?

- I think refactoring is important because it makes the playbook shorter, clear and easier to understand. Refactoring also can make the playbook run faster because it reduces the number of tasks that ansible needs to perform.

2. When do we use the “when” command in playbook?

- We will use the when command when we want a task to run only if a certain condition is true. For example, we can use it to run apt tasks only on ubuntu and dnf tasks only on the centOS.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of code and concept , and code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use this tool to help me identify and know what each component does in the structure of the yaml playbook. I also ask help for a bit to assist me when I don’t know a part of code in the playbook**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.’s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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